Technical supplement: Modeling Countermovement-Jump Recovery

All values are read from the public fixed-mode reference-study receipt and the public model-qualification summary. Descriptive synthetic results; not athlete accuracy.

S1 All evaluated variants: mean six-cell NRMSE across three roots

Variant	Mean	SD	Min	Max
ridge_full	0.736270	0.003349	0.732724	0.739378
ridge_n24	0.736889	0.003809	0.733212	0.740818
ridge_n48	0.736602	0.003261	0.733219	0.739726
ridge_n96	0.736270	0.003349	0.732724	0.739378
ridge_no_current_exposure	0.867735	0.006220	0.861227	0.873620
ridge_no_participant_context	0.736346	0.003327	0.732837	0.739455
ridge_no_prior_history	0.819681	0.005899	0.812906	0.823679
structured_full	0.773316	0.005490	0.767583	0.778526
structured_n24	0.773624	0.005429	0.767939	0.778755
structured_n48	0.773621	0.005475	0.767828	0.778711
structured_n96	0.773316	0.005490	0.767583	0.778526
structured_no_prior_history	0.820342	0.005741	0.813749	0.824241
training_constant	1.036038	0.007308	1.027789	1.041703
training_constant_n24	1.038164	0.007410	1.029805	1.043927
training_constant_n48	1.037354	0.007386	1.029019	1.043086
training_constant_n96	1.036038	0.007308	1.027789	1.041703
zero_innovation	2.430659	0.008127	2.421275	2.435478

S2 Per-cell NRMSE by root (four reference models)

Model	Cell	Root A	Root B	Root C	Mean
zero_innovation	H24/net_impulse_innovation	2.7578	2.7595	2.7679	2.7618
zero_innovation	H24/mean_force_innovation[legacy key]	3.0431	3.0492	3.0084	3.0335
zero_innovation	H48/net_impulse_innovation	2.2925	2.2226	2.2780	2.2644
zero_innovation	H48/mean_force_innovation[legacy key]	2.5260	2.5509	2.5492	2.5420
zero_innovation	H72/net_impulse_innovation	1.8829	1.8443	1.8925	1.8732
zero_innovation	H72/mean_force_innovation[legacy key]	2.1106	2.1012	2.1153	2.1090
training_constant	H24/net_impulse_innovation	1.0534	1.0364	1.0456	1.0451
training_constant	H24/mean_force_innovation[legacy key]	1.0548	1.0408	1.0474	1.0477
training_constant	H48/net_impulse_innovation	1.0359	1.0225	1.0369	1.0318
training_constant	H48/mean_force_innovation[legacy key]	1.0447	1.0284	1.0452	1.0395
training_constant	H72/net_impulse_innovation	1.0286	1.0189	1.0270	1.0248
training_constant	H72/mean_force_innovation[legacy key]	1.0329	1.0197	1.0297	1.0274
ridge_full	H24/net_impulse_innovation	0.6513	0.6511	0.6541	0.6522
ridge_full	H24/mean_force_innovation[legacy key]	0.6303	0.6301	0.6128	0.6244
ridge_full	H48/net_impulse_innovation	0.7522	0.7659	0.7614	0.7598
ridge_full	H48/mean_force_innovation[legacy key]	0.7371	0.7379	0.7208	0.7319
ridge_full	H72/net_impulse_innovation	0.8343	0.8296	0.8369	0.8336
ridge_full	H72/mean_force_innovation[legacy key]	0.8149	0.8217	0.8104	0.8157
structured_full	H24/net_impulse_innovation	0.6942	0.6983	0.6955	0.6960
structured_full	H24/mean_force_innovation[legacy key]	0.6633	0.6649	0.6444	0.6575
structured_full	H48/net_impulse_innovation	0.7972	0.8125	0.8041	0.8046
structured_full	H48/mean_force_innovation[legacy key]	0.7678	0.7690	0.7463	0.7610
structured_full	H72/net_impulse_innovation	0.8773	0.8764	0.8800	0.8779
structured_full	H72/mean_force_innovation[legacy key]	0.8432	0.8501	0.8352	0.8429

S3 Recorded camp-bootstrap intervals (all 17 variants; difference is from zero innovation)

Root	Variant	NRMSE 95% interval	Difference from zero
A	ridge_full	[0.7280, 0.7462]	[-1.7522, -1.6450]
A	ridge_n24	[0.7275, 0.7465]	[-1.7521, -1.6449]
A	ridge_n48	[0.7284, 0.7462]	[-1.7521, -1.6448]
A	ridge_n96	[0.7280, 0.7462]	[-1.7522, -1.6450]
A	ridge_no_current_exposure	[0.8590, 0.8780]	[-1.6179, -1.5157]
A	ridge_no_participant_context	[0.7281, 0.7462]	[-1.7522, -1.6449]
A	ridge_no_prior_history	[0.8108, 0.8339]	[-1.6662, -1.5588]
A	structured_full	[0.7638, 0.7843]	[-1.7147, -1.6081]
A	structured_n24	[0.7636, 0.7851]	[-1.7139, -1.6078]
A	structured_n48	[0.7643, 0.7848]	[-1.7140, -1.6077]
A	structured_n96	[0.7638, 0.7843]	[-1.7147, -1.6081]
A	structured_no_prior_history	[0.8117, 0.8341]	[-1.6659, -1.5582]

Root	Variant	NRMSE 95% interval	Difference from zero
A	training_constant	[1.0316, 1.0535]	[-1.4437, -1.3438]
A	training_constant_n24	[1.0336, 1.0562]	[-1.4414, -1.3415]
A	training_constant_n48	[1.0328, 1.0552]	[-1.4423, -1.3424]
A	training_constant_n96	[1.0316, 1.0535]	[-1.4437, -1.3438]
A	zero_innovation	[2.3773, 2.4947]	[0.0000, 0.0000]
B	ridge_full	[0.7338, 0.7453]	[-1.7156, -1.6484]
B	ridge_n24	[0.7349, 0.7470]	[-1.7140, -1.6477]
B	ridge_n48	[0.7342, 0.7456]	[-1.7152, -1.6484]
B	ridge_n96	[0.7338, 0.7453]	[-1.7156, -1.6484]
B	ridge_no_current_exposure	[0.8530, 0.8699]	[-1.5906, -1.5298]
B	ridge_no_participant_context	[0.7340, 0.7454]	[-1.7154, -1.6484]
B	ridge_no_prior_history	[0.8169, 0.8320]	[-1.6330, -1.5635]
B	structured_full	[0.7735, 0.7846]	[-1.6766, -1.6092]
B	structured_n24	[0.7735, 0.7848]	[-1.6765, -1.6091]
B	structured_n48	[0.7735, 0.7847]	[-1.6765, -1.6089]
B	structured_n96	[0.7735, 0.7846]	[-1.6766, -1.6092]
B	structured_no_prior_history	[0.8174, 0.8326]	[-1.6326, -1.5626]
B	training_constant	[1.0216, 1.0348]	[-1.4222, -1.3650]
B	training_constant_n24	[1.0235, 1.0369]	[-1.4201, -1.3630]
B	training_constant_n48	[1.0228, 1.0361]	[-1.4210, -1.3638]
B	training_constant_n96	[1.0216, 1.0348]	[-1.4222, -1.3650]
B	zero_innovation	[2.3887, 2.4554]	[0.0000, 0.0000]
C	ridge_full	[0.7264, 0.7401]	[-1.7338, -1.6691]
C	ridge_n24	[0.7264, 0.7412]	[-1.7329, -1.6687]
C	ridge_n48	[0.7265, 0.7412]	[-1.7330, -1.6686]
C	ridge_n96	[0.7264, 0.7401]	[-1.7338, -1.6691]
C	ridge_no_current_exposure	[0.8649, 0.8821]	[-1.5913, -1.5304]
C	ridge_no_participant_context	[0.7265, 0.7402]	[-1.7337, -1.6690]
C	ridge_no_prior_history	[0.8051, 0.8216]	[-1.6548, -1.5869]
C	structured_full	[0.7612, 0.7751]	[-1.7005, -1.6325]
C	structured_n24	[0.7614, 0.7755]	[-1.6999, -1.6324]
C	structured_n48	[0.7612, 0.7755]	[-1.7000, -1.6324]
C	structured_n96	[0.7612, 0.7751]	[-1.7005, -1.6325]
C	structured_no_prior_history	[0.8061, 0.8222]	[-1.6540, -1.5859]
C	training_constant	[1.0292, 1.0493]	[-1.4234, -1.3686]
C	training_constant_n24	[1.0311, 1.0517]	[-1.4212, -1.3665]
C	training_constant_n48	[1.0304, 1.0509]	[-1.4220, -1.3672]
C	training_constant_n96	[1.0292, 1.0493]	[-1.4234, -1.3686]
C	zero_innovation	[2.3994, 2.4699]	[0.0000, 0.0000]

S4 Original architecture routes (descriptive, no performance)

Family	Repeat	Parameters	CPU vs original source	Saved CUDA arrays
M0	r0	1346	Exact source parity	Within recorded tolerance
M0	r1	1346	Exact source parity	Within recorded tolerance
M1	r0	2690	Exact source parity	Within recorded tolerance
M1	r1	2690	Exact source parity	Within recorded tolerance
M2	r0	6850	Exact source parity	Within recorded tolerance
M2	r1	6850	Exact source parity	Within recorded tolerance
M3	r0	7138	Exact source parity	Within recorded tolerance
M3	r1	7138	Exact source parity	Within recorded tolerance
M4	r0	10626	Exact source parity	Outside recorded tolerance
M4	r1	10626	Exact source parity	Outside recorded tolerance
M5	r0	10658	Exact source parity	Outside recorded tolerance
M5	r1	10658	Exact source parity	Outside recorded tolerance
M6	r0	10738	Exact source parity	Outside recorded tolerance
M6	r1	10738	Exact source parity	Outside recorded tolerance
M7	r0	10786	Exact source parity	Outside recorded tolerance
M7	r1	10786	Exact source parity	Outside recorded tolerance
M8	r0	10796	Exact source parity	Outside recorded tolerance
M8	r1	10796	Exact source parity	Outside recorded tolerance